MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD

ELECTRICAL ENGINEERING DEPARTMENT

Class Test-1 (Session 2025-26) (SET-1)

Course: B. Tech
Semester: 1^{st}
Section: E, F, G, H

Subject: Fundamentals of Electrical Engineering
Time: 60 Minutes

Subject Code: BEE-101
Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO2	CO2	CO2	CO1	CO1	CO3
Bloom's Level	K1	K3	K2	K3	K3	K2

Note: 1) This paper contains three sections, Section (A) (B) & (C) 2) All sections are compulsory

Section (A)

Attempt this question. This question carries 2 marks. (1*2 = 2)

1) **Derive expression for resonant frequency in series R-L-C circuit.**

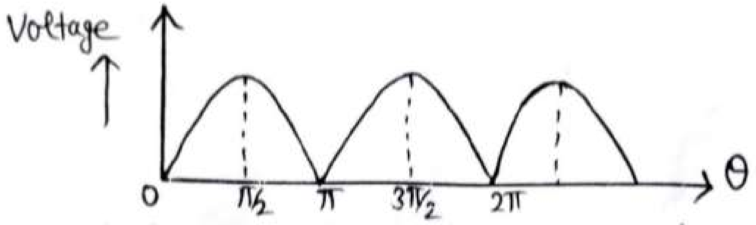
Section (B)

Attempt all questions. Each question carries 3 marks. (2*3 = 6)

2) **An alternating voltage is given by $v = 141.4 \sin 314t$. Find (i) the instantaneous value of voltage when 't' is 3 ms (ii) the time taken for the voltage to reach 100 V for the first time after passing through zero value.**

OR

Find rms average value and peak factor of the voltage waveform shown in figure.



3) **For an ac circuit $v = 200 \sin 377t$ volts, and $i = 8 \sin (377t - 30°)$ amperes. Determine true power, apparent power and reactive power of circuit.**

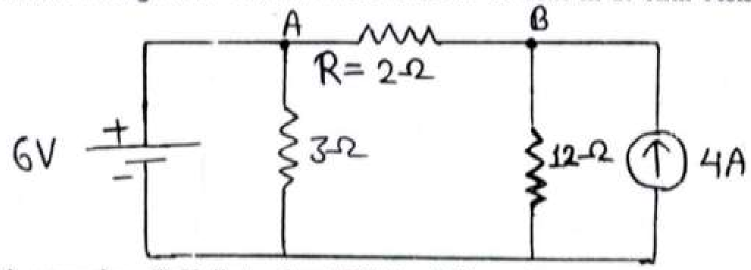
OR

Show that the average power for purely inductive circuit is zero.

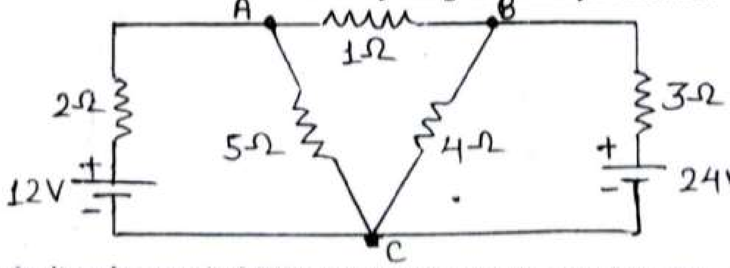
Section (C)

Attempt all questions. Each question carries 4 marks. (3*4 = 12)

4) **State Kirchhoff voltage law. Calculate the value of current in R ohm resistance by using mesh analysis method.**



5) **Define with examples (i) Unilateral and Bilateral Elements.**
(ii) Determine the value of current in each branch by using nodal analysis method.



6) **(i) Give the names (write only names) of different types of magnetic materials with examples.**
(ii) Derive emf equation of normal two winding transformer.

For any query contact to: Mr Saurabh Saxena 9456031300/7895043195, Mr. Tanuj Kr. Bisht 9149273074

approved

08-10-25

Pooja (21)

MORADABAD INSTITUTE OF TECHNOLOGY, MORADABAD

ELECTRICAL ENGINEERING DEPARTMENT

Class Test-1 (Session 2024-25) (SET-1)

Course: B. Tech
Subject: Fundamentals of Electrical Engineering
Subject Code: BEE-201

Semester: 1st

Section: A, B, C, D
Time: 60 Minutes
Max. Marks: 20

S. No.	1	2	3	4	5	6
CO No.	CO3	CO2	CO2	CO1	CO1	CO2
Bloom's Level	K1	K2	K3	K3	K3	K4

Note: 1) This paper contains three sections, Section (A) (B) & (C) **2)All sections are compulsory**

Section (A)

Attempt this question. This question carries 2 marks. (1*2 = 2)

1) With the help of examples classified different types of magnetic materials.

Section (B)

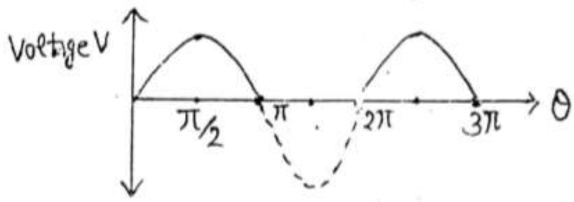
Attempt all questions. Each question carries 3 marks. (2*3 = 6)

2) Derive an expression for form factor and peak factor for sinusoidal voltage.

OR

Find rms, average value and peak factor of the voltage waveform shown in figure.

Peak factor = $\frac{\text{Max}}{\text{Rms}}$



3) The voltage applied to a circuit is $v = 100 \sin(\omega t + 30^0)$ & current flowing in the circuit is $i = 20 \sin(\omega t + 60^0)$. Determine the impedance, resistance, reactance, power and power factor of the circuit.

OR

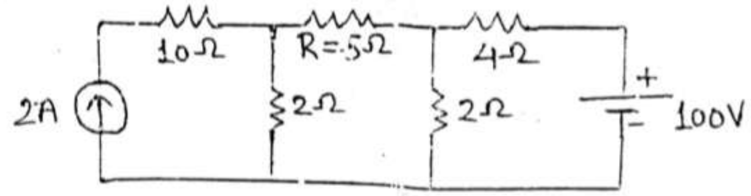
An alternating voltage is given by $v = 141.4 \sin 314\, t$. Find (i) frequency (ii) rms value (iii) average value (iv) the instantaneous value of voltage when 't' is 3 ms (v) the time taken for the voltage to reach 100 V for the first time after passing through zero value.

Section (C)

Attempt all questions. Each question carries 4 marks. (3*4 = 12)

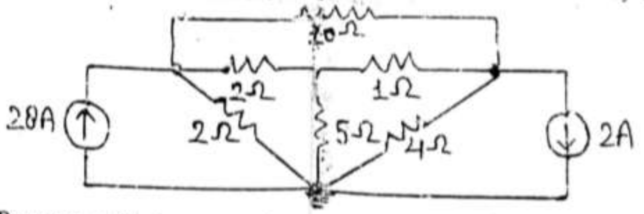
4) (a) Define with examples (i) Unilateral and Bilateral Elements (ii) Active and Passive Elements

(b) State Kirchhoff voltage law. Calculate the value of current in R ohm resistance by using mesh analysis method.



Ferrous
nickel
cobalt

5) State Kirchhoff current law. Determine the value of current in each branch by using nodal analysis method.



6) Define the term Resonance. Derive expression for resonant frequency in series R-L-C circuit.Also draw resonance curve.

For any query contact to: Mr.Saurabh Saxena 789504315, Mr. Akash Kr. Gautam 8954751891, Mr. Lalit Parchha 7060460472

approved Saxena 17.05.25